Seminar1

luni, 11 octombrie 2021 10:07

- limbaj = mulțime finită sau infinită de cuvinte
  vidă / nevidă

$\emptyset = \{\}$

- alfabet = mulțime finită de simboluri
- cuvânt = secvență finită de simboluri

laptop.

$l = [$ "laptop", "dog", "cat" $]$

$\{ a, b, c, d, \ldots\ldots, z \}$

mama
joacă

{ mama, joacă, monitor, telefon... }

L_1, L_2 două limbaje

$L_1 \cup L_2$, $L_1 \cap L_2$, $L_1 \setminus L_2$, $\complement L_1$, $\complement L_2$,

$L_1 \circ L_2$ concatenare

L_1^*, L_2^* Kleene Star

alfabet
$\Sigma = \{0, 1\}$

Σ^* = toate cuvintele care se pot forma cu simbolurile din Σ

w = cuvânt

$L_1 = \{ 01, 110 \}$

$L_2 = \{ 1, 11, 000, 01 \}$

$L_1 \cup L_2 = \{ w \in \Sigma^* / w \in L_1 \text{ sau } w \in L_2 \}$

$= \{ 01, 110, 1, 11, 000 \}$ $\quad \varepsilon$ = șirul vid

$L_1 \circ L_2 = \{ w_1 w_2 \ / \ w_1 \in L_1 \text{ și } w_2 \in L_2 \}$

$= \{ 011, 0111, 01000, 0101, 1101, 11011, 110000, 11001 \}$

0101

$L_1 = \{ 01, 0 \}$

$L_2 = \{ 11, 1 \}$

$\{ 0111, 011, 01 \}$

$L_1 = \{ 01, 0 \}$

$L_1^* = L_1^0 \cup L_1^1 \cup L_1^2 \cup L_1^3 \cup \ldots\ldots$

$= \{ \varepsilon \cup \{01, 0\} \cup L_1 \circ L_1 \cup L_1 \circ L_1 \circ L_1 \ldots\ldots$

$= \{e, 01, 0, 0101, 010, 001, 00\}$

$L_1 \cdot L_1 = 0101 \quad 010 \quad 001 \quad 00$

$L_1 = \emptyset$

$L_1^* = L_1^0 \cup L_1^1 \cup L_1^2 \cup L_1^3 \cup \dots$

$= \emptyset^0 \cup \emptyset \cup \emptyset^2 \cup \emptyset^3$

$= \{e\}$

$\emptyset^* = \{e\}$ " " + " " + " " = " "

$e \cdot e \cdot e = e$

$L = \{e\}$

$L^* = L^0 \cup L^1 \cup L^2 \cup \dots$

$= e \cup e \cup$

$= \{e\}$

$E = \emptyset$ ←

$E = e$ ←

$E = a$

$\Sigma = \{a, b\}$

$E = a$

dc. α, β sunt E.R. at. $\alpha \cup \beta$, $\alpha \circ \beta$, α^*, β^* sunt E.R.

$\Sigma = \{0, b\}$

$(ab \cup e)^*$ $\qquad$ $(ab \cap e)^*$

$\Sigma = \{a, b\}$

$(ab \cup ba)(abb \cup e)^*$

$= (ab \cup b^0) \cdot e \cdot (abb \cup e)$

$ab \cdot e = ab$

$ba \cdot e \cdot abb = ba$

$ab < e \cdot abb = ab \cdot abb$

$ba < abb \cdot e = ab \cdot babb$

$= (ab \cup a)^* \cdot b$

$(abb \cup b)$

$ba \cdot \varepsilon_{abb} = ba$
$ab < \varepsilon_{abb} = ab$
$ba < \varepsilon_{abb} = ab \cdot abb \cdot ba$

1. $(abb \cup \varepsilon)(abb \cup \varepsilon)$
2. $(abb \cup \varepsilon)(abb \cup \varepsilon)(abb \cup \varepsilon)$

$(ab \cup ba)(abb \cup \varepsilon)^*$

$\{ab, ba, abb\} = L \cdot L^*$

$L^* = \{\varepsilon, ab, ba, abb, \ldots\}$

$((ab \cup ba) \cdot (abb \cup \varepsilon))^*$

$ab \cdot abb$

I. $ab = \rightsquigarrow ab$

II. ba

1. $*$
2. $\cdot$
3. $\cup$

$(a \cup b)^*$

$(ab)^*$

$L_1 = \{01, 11\}$ $\quad L_1 = \{01\} \cup \{11\}$

$E_1 = 01 \cup 11 \rightarrow \{01, 11\}$

$E_2 = (0 \cup 1)1 \rightarrow 0 \cdot 1 = 01$ $\quad \{01, 11\}$
$1 \cdot 1 = 11$

$E_3 = (0 \cup 1\varepsilon)1$

$(0 \cup \varepsilon) \cdot 1$

$E_1 = 01 \cup 11$ $\quad E_1 = E_{10} \cdot E_{11} \cup E_{11} \cdot E_{11}$

$E_{10} = 0$

$E_{11} = 1$

$L \neq E$

$L = \mathcal{L}(E)$

2. $\Sigma = \{0, 1\}$

$L = \{ w \in \Sigma^* / w \text{ începe cu } 00 \}$

$E = 00 \cup \Sigma^*$ $\quad 00 < \Sigma^*$ $\quad 1100$

$E = 00(0 \cup 1)^*$

$\mathcal{L}(E)$. $00, e, 0, 1, 00, 01, 10, 11 \ldots$

$L \subset \mathcal{L}(E)$

$M_1 \cup M_2$

3. $\Sigma = \{x, y, z\}$

$L = \{ w \in \Sigma^* / w \text{ începe și se termină cu același simbol} \}$

$E = x(x \cup y \cup z)^* x \cup y(x \cup y \cup z)^* y \cup z(x \cup y \cup z)^* z \cup e$

$\cup x \cup y \cup z$

$E = (x \cup y \cup z)(x \cup y \cup z)^*(x \cup y \cup z)$

$e \quad (x \cup y \cup z)^* x \quad (x \cup y \cup z)^* x (x \cup y \cup z)^* x \ldots$

$E = x(x \cup y \cup z)^* x \cup$

$y \quad y \quad \cup$

$z \quad z$

$\Sigma^* \quad (x \cup y \cup z)^*$

$\Sigma^* \quad (0 \cup 1)^*$

$\Sigma = \{0, 1\}$

$\Sigma^* = \{ e, 0, 1, 00, 01, 10, 11, \ldots \}$

$\Sigma^* = \{ e, 0, 1, 00, 01, 10, 11, \ldots \}$

$\Sigma \cdot \Sigma = \{0,1\} \cdot \{0,1\}$

$\Sigma\Sigma\Sigma = \{0,1\}\{0,1\}\{0,1\} = \{00, 01, 10, 11\}\{0,1\}$

$= \{000, 001, 010, 011, 100, 101, 110, 111\}$

$(0 \cup 1)^* = (0 \cup 1)^0 \cup (0 \cup 1)^1 \cup (0 \cup 1)^2 \ldots$

$e, \quad 0, 1, \quad 00, 01, 10, 11,$

4. $\Sigma = \{a, b\}$

$L = \{ w \in \Sigma^* / \; w \text{ are lungime pară} \}$

$\#(w) \; \vdots \; 2$

$E_1 = (a \cup b)(ab \cup ba)^*(a \cup b)$ ✗

$aaaa \; bbbb$ $\quad baab$

$E_2 = ((a \cup b)(a \cup b))^*$ ✓

$E_3 = (aa \cup bb \cup ba \cup ab)^*$ — ✓

$E_3 \setminus e$ ✗

$\mathcal{L}(E_3) = L_3.$

$L_3 \setminus \{e\}$ ✓.

$\Sigma = \{0, 1\}$

5. $L = \{ w \in \Sigma^* / \#_0(w) \geq 2$ y w termina en $1 \}$

6. $L = \{ w \in \Sigma^* / \#_{01}(w) = 0 \}$

7. $L = \{ w \in \Sigma^* /$ w empieza con nro. par de 1 y x termina con nro. par de $0 \}$

8. $L = \{ w \in \Sigma^* / \#_1(w) \leq 3 \}$

9. $L = \{ w \in \Sigma^* / \#_{01}(w) = \#_{10}(w) \}$